

Gannaju Sampath Chary

Python AI Engineer | Data Scientist | Computer Vision | FastAPI | TensorFlow | PostgreSQL

<http://localhost:3000> | [RAISE Story](#) | [RAISE App](#) | [Projects UI](#)

Hyderabad, Telangana, India
gsampathchary454@gmail.com
[linkedin.com/in/sampath-chary-3346a7233](https://www.linkedin.com/in/sampath-chary-3346a7233)
github.com/GSampathChary

PROFESSIONAL SUMMARY

Python AI Engineer with 1.10+ years of hands-on experience building machine learning, computer vision, mobile, web, and backend systems. Experienced across Python, TypeScript, SQL, TensorFlow, PyTorch, OpenCV, FastAPI, Spring Boot, PostgreSQL, TensorFlow Lite, and ONNX Runtime, with practical ownership across data preparation, exploratory analysis, feature engineering, model training, evaluation, deployment, and production support.

CORE COMPETENCIES

AI / Machine Learning	Backend & APIs
Computer Vision, model training, deployment, optimization	Spring Boot, FastAPI, REST APIs, microservices
Full Stack	Data & Delivery
Flutter, React.js, Next.js, UI implementation	PostgreSQL, dataset prep, release support

TECHNICAL SKILLS

Languages	Python, SQL, JavaScript, Java, Dart
AI / ML	TensorFlow, PyTorch, Scikit-learn, Keras, ONNX Runtime, TensorFlow Lite, OpenCV
Data Science & BI	Pandas, NumPy, Power BI, DAX, Power Query, Exploratory Data Analysis, Feature Engineering, Model Evaluation
Backend & APIs	FastAPI, Spring Boot, REST APIs, Microservices
Frontend & Apps	Flutter, React.js, Next.js, HTML5, CSS3
Database & Delivery	PostgreSQL, MySQL, Firebase, Docker, Google Colab, Google Play Console
Tools	Git, GitHub, Postman, VS Code, Linux

PROFESSIONAL EXPERIENCE

Young Professional - I (AI & Full Stack Developer) ICAR - Indian Institute of Rice Research (IIRR), Hyderabad December 2024 - Present	AI-powered mobile, web, and backend applications for agricultural research.
---	---

- Develop and maintain Python-based AI, web, and backend applications for agricultural research workflows.
- Build production mobile experiences with Flutter and web experiences with React.js and Next.js.

- Design FastAPI and Spring Boot services for authentication, predictions, feedback, and workflow orchestration.
- Collect, clean, preprocess, annotate, and augment domain datasets for model development and validation.
- Perform exploratory data analysis, feature engineering, and dataset quality checks before training.
- Train, validate, and optimize deep learning models for image classification and computer vision use cases.
- Integrate AI models using TensorFlow, PyTorch, TFLite, and ONNX Runtime for mobile and server inference.
- Design PostgreSQL data models and role-based workflows for scientists, admins, and end users.
- Collaborate with subject-matter experts for ground-truth validation, evaluation, and iteration.
- Deploy applications and backend services and support production releases and debugging.

FLAGSHIP PROJECT - RAISE

RAISE

Rice AI Stress Evaluator
Project Story
Open App

Flutter | TensorFlow | Spring Boot | Python | PostgreSQL | TFLite | ONNX Runtime

AI-powered image-based platform for identifying biotic and abiotic rice crop stresses, combining mobile inference, server-side AI services, scientist validation, and structured agricultural data management.

- Developed the Flutter Android application with image capture, crop-stress scanning, prediction history, offline inference, multilingual support, and farmer-focused workflows.
- Built the React.js scientist/admin portal and Spring Boot backend with REST APIs for authentication, predictions, expert review, data management, and administration.
- Integrated FastAPI-based Python AI services with TensorFlow/PyTorch models for server-side image inference.
- Trained and evaluated image-classification models including EfficientNetV2, EfficientNetV1, ResNet, MobileNet, and InceptionNet; explored YOLOv5 and YOLOv8 for computer-vision workflows.
- Converted and optimized models for lightweight inference using TensorFlow Lite and ONNX Runtime.
- Implemented Farmer, Scientist, and Admin roles, including expert review for visually confusing cases and feedback-driven model improvement.
- Designed PostgreSQL data models for users, predictions, review workflows, and application history.

PERSONAL PROJECTS

InsightBI AI

India-focused Power BI and AI analytics platform
Project Story

Power BI | DAX | Power Query | PostgreSQL | Python | FastAPI | Pandas | Scikit-learn | Next.js | React
Open live experience
No backend link

End-to-end business intelligence platform that transforms sales data into Power BI-ready analytics, executive dashboards, customer RFM segments, forecasting, anomaly detection, and AI-assisted analysis. Built with a Kimball-style PostgreSQL star schema, validated ETL pipelines, FastAPI reporting APIs, and India-ready INR formatting and regional demo data.

VistaraAI

AI-powered PVC interior design assistant
Project Story

Next.js | TypeScript | React | FastAPI | Python | LangChain | LangGraph | PostgreSQL
Open live experience
Backend API

Production-ready full-stack AI platform for PVC kitchen cupboards, wardrobes, TV units, and custom interiors. Combines grounded knowledge retrieval, assistant workflows, computer vision insights, geo-targeted SEO, and lead capture support.

BharatEye India-focused GeoAI and Earth Observation platform Project Story	React Vite FastAPI Deck.gl Mapbox GL JS Zustand ECharts Open live experience Backend API India-focused GeoAI and Earth Observation platform that visualizes simulated satellite intelligence, disaster monitoring, AOI analysis, maritime surveillance, crop stress, and infrastructure risk through interactive geospatial dashboards and live WebSocket-powered updates.
RAISE Rice AI Stress Evaluator Project Story	Flutter TensorFlow Spring Boot Python PostgreSQL TFLite ONNX Runtime Open live experience No backend link Production AI platform for identifying biotic and abiotic rice crop stresses using image-based diagnosis. Includes mobile AI inference, scientist/admin portal, expert validation, role-based access, prediction history, multilingual UI, and offline inference.
RiceGPT AI Agricultural AI assistant concept Project Story	Next.js TypeScript React Open live experience Backend API Domain AI assistant concept for agricultural and rice-related queries, pairing a structured knowledge base with a public web deployment.
ResumeAI Pro AI resume optimization concept Project Story	Next.js TypeScript React Tailwind CSS FastAPI Open live experience Backend API AI-assisted resume analysis and refinement concept focused on recruiter-friendly guidance and practical deployment.
Interview Copilot AI Interview preparation companion Project Story	React TypeScript Prompt Engineering Open live experience Backend API Interview preparation companion for technical question practice, answer structuring, and confidence building.
DataInsight AI AI data analysis concept Project Story	Python Next.js TypeScript Data Visualization Open live experience Backend API AI-assisted exploratory data analysis concept for pattern discovery, summary reporting, and decision support.
AutoML Studio Automated ML workflow concept Project Story	Python FastAPI Scikit-learn Next.js Gemini Open live experience Backend API End-to-end automated machine learning concept for dataset upload, preprocessing, training, comparison, and evaluation.

ADDITIONAL PROJECT EXPERIENCE

Rice Clinic

Flutter | Localization | Image Matching | Google Apps Script | Google Sheets

Multilingual rice advisory application for diseases, pests, weeds, nutrient deficiencies, cultivation practices, and fertilizer/spray recommendations.

KEY ACHIEVEMENTS

- Delivered an end-to-end AI application spanning dataset preparation, model training, mobile/web development, backend APIs, database design, deployment, and maintenance.
- Published the RAISE IIRR Android application on the Google Play Store.
- Built a multi-role AI platform connecting farmers, scientists, administrators, and expert reviewers through a single application ecosystem.
- Developed model-deployment workflows supporting both offline mobile inference and online server inference.
- Built reusable backend and AI-service architecture using Spring Boot, FastAPI, PostgreSQL, and Python.

EDUCATION

B.Tech, Computer Science

2019 - 2023

Gurunanak Institutions Technical Campus, Hyderabad, Telangana

University not explicitly specified in the current resume source.

Portfolio: <http://localhost:3000> | Projects: </projects> | Experience: </experience> | Resume: </resume/resume.pdf>